

SQL WHERE Clause

Operators and Keywords - Quick Reference

The WHERE clause filters rows so only records matching a condition are returned.

```
SELECT column_list FROM table_name WHERE condition;
```

Sample table: students(id, name, age, city, marks)

1. Comparison Operators

Operator	Use	Example
=	Equal to	WHERE age = 20
<> or !=	Not equal	WHERE city <> 'Delhi'
>	Greater than	WHERE marks > 80
<	Less than	WHERE age < 18
>=	Greater than or equal	WHERE marks >= 50
<=	Less than or equal	WHERE age <= 25

2. Logical Operators

Keyword	Use	Example
AND	Both conditions must be true	WHERE age > 18 AND city = 'Kochi'
OR	Any one condition true	WHERE city = 'Kochi' OR city = 'Tirur'
NOT	Reverses a condition	WHERE NOT city = 'Delhi'

3. LIKE (Pattern Matching)

Used with text. It uses two wildcards: % = any number of characters (zero, one or many), and _ = exactly one character.

Pattern	Meaning	Example matches
LIKE 's%'	Starts with s	Sam, Sara
LIKE '%s'	Ends with s	Ramesh, Anas
LIKE '%s%'	Contains s anywhere	Rasheed, Sam
LIKE '_a%'	Second letter is a	Ravi, Kavya
LIKE 'A____'	Starts with A, exactly 4 letters	Anil, Arun
LIKE '%an'	Ends with 'an'	Rehman, Kiran

```
-- Names that end with s
SELECT * FROM students WHERE name LIKE '%s';
```

```
-- Names that start with A
SELECT * FROM students WHERE name LIKE 'A%';
```

```
-- Names that contain "an"
SELECT * FROM students WHERE name LIKE '%an%';
```

```
-- NOT LIKE gives the opposite
```

```
SELECT * FROM students WHERE name NOT LIKE '%s';
```

4. IN (match any value in a list)

```
SELECT * FROM students WHERE city IN ('Kochi', 'Tirur', 'Calicut');
```

Same as writing multiple OR conditions. NOT IN gives the opposite.

5. BETWEEN (range, both ends included)

```
SELECT * FROM students WHERE marks BETWEEN 60 AND 90;  
SELECT * FROM students WHERE age NOT BETWEEN 18 AND 25;
```

6. IS NULL / IS NOT NULL

Checks for empty (missing) values. Do not use = NULL, it will not work.

```
SELECT * FROM students WHERE city IS NULL;  
SELECT * FROM students WHERE city IS NOT NULL;
```

7. EXISTS

True if a subquery returns at least one row.

```
SELECT * FROM students s  
WHERE EXISTS (SELECT 1 FROM results r WHERE r.id = s.id);
```

8. ANY / ALL

Compare a value with a subquery result.

```
-- greater than at least one value  
SELECT * FROM students  
WHERE marks > ANY (SELECT marks FROM students WHERE city = 'Kochi');  
  
-- greater than every value  
SELECT * FROM students  
WHERE marks > ALL (SELECT marks FROM students WHERE city = 'Kochi');
```

Quick Summary

Keyword	Purpose
=, <>, >, <, >=, <=	Compare values
AND, OR, NOT	Combine conditions
LIKE	Pattern search (%, _)
IN	Match a list of values
BETWEEN	Match a range
IS NULL	Check empty values
EXISTS	Check if subquery has rows
ANY / ALL	Compare with subquery values

Tip: LIKE is case-insensitive in MySQL by default, but case-sensitive in PostgreSQL (use ILIKE there for case-insensitive matching).